$$\chi^0{}_1(x) = \chi^1{}_0(x) = \frac{\Delta}{a^4r^2\left(\Delta - a^2\right) + a^2r^3\left(4Mr\left(M-r\right) + 2r^3 - Q^2\left(a^2+2Mr-r^2\right) + \Delta(2M-r)\right) + r^6\left(\Delta - a^2\right)}$$
$$\Bigg[\left(2a^2r^2+r^4\right)\cosh(\zeta)\left(Q^2-Mr\right)\sqrt{\frac{\left(a^2+r^2\right)^2-a^2\Delta}{\left(\left(a^2+r^2\right)^2-a^2\Delta\right)\left(r(r-2M)+Q^2\right)+a^2\left(Q^2-2Mr\right)^2}}-$$
$$\frac{1}{\left(\left(a^2+r^2\right)^2-a^2\Delta\right)^{3/2}}\Bigg[a\Big(a^4\left(-\left(r(r-3M)+Q^2\right)\right)+a^2\Big(Q^2\left(a^2+3Mr+Q^2+r^2\right)+r\big(\Delta(r-3M)-$$
$$r\left(6M^2-3Mr+r^2\right)\big)-Q^4\Big)+r^4\left(r(r-5M)+3Q^2\right)\Big)$$
$$\left(\frac{a\cosh(\zeta)\left(\left(a^2+r^2\right)^2-a^2\Delta\right)\left(2Mr-Q^2\right)}{\sqrt{\left(\left(a^2+r^2\right)^2-a^2\Delta\right)\left(r(r-2M)+Q^2\right)+a^2\left(Q^2-2Mr\right)^2}}+\sinh(\zeta)\left(a^2\left(2Mr-Q^2+r^2\right)+r^4\right)\right)\Bigg]\Bigg] \tag{15}$$

$$\chi^1{}_3(x) = -\chi^3{}_1(x)\frac{\sqrt{\Delta}}{r^4}\Bigg[\cosh(\zeta)\left(\frac{a\left(2Mr-Q^2\right)\left(2a^2\left(Q^2-Mr\right)+r^4\right)}{\sqrt{\left(a^2+r^2\right)^2-a^2\delta}}+2a\left(Mr-Q^2\right)\right)$$
$$\sqrt{\frac{\left(a^2+r^2\right)^2-a^2\Delta}{\left(\left(a^2+r^2\right)^2-a^2\delta\right)\left(-2Mr+Q^2+r^2\right)+a^2\left(Q^2-2Mr\right)^2}}+\sinh(\zeta)\left(a^2\left(2Mr-Q^2+r^2\right)+r^4\right)\Bigg] \tag{16}$$

The LLT is defined by Eq.(14), where m is taken to be the mass of the particle species under consideration. This is an infinitesimal LLT since $\lambda_{ab} = -\lambda_{ba}$. The LLT then turns out to have only four nonvanishing terms which can be separated in to two symmetries,

$$\begin{aligned}\lambda^0{}_1(x) =& \lambda^1{}_0(x)\\ \lambda^1{}_3(x) =& -\lambda^3{}_1(x)\end{aligned} \tag{17}$$

The explicit expressions are lengthy and so are represented graphically in Fig.(1) for some average parameters of a Kerr-Newman black hole.

This turns out to also be a boost along the 1-axis and a rotation about the 2-axis, as in Schwarzschild black hole case. With constant momentum $p^a(x) = (m\cosh(\zeta), 0, 0, m\sinh(\zeta))$ pointing in the 3-axis, the change of the spin becomes a rotation as follows.

After an infinitesimal proper time $d\tau$, the particle moves in the 3-axis by an amount, $\delta\phi = u^\varphi d\tau$. Over this the momentum in the local inertial frame transforms under the LLT, $\Lambda^a{}_b(x) = \delta^a{}_b + \lambda^a{}_b(x)\,d\tau$, which corresponds to a unitary operator that acts on the state of the particle. This operator changes the spin, in particular it acts like the unitary matrix U in

$$\begin{aligned}&U\left(\Lambda(x)\right)\left|p^a(x), \sigma; x\right\rangle = \\ &\qquad\sum_{\sigma'} D^{(1/2)}_{\sigma'\sigma}\left(W(x)\right)\left|\Lambda p^a(x), \sigma'; x\right\rangle,\end{aligned} \tag{18}$$

as in [9], where $W(x) = W(\Lambda(x), p(x)) = L^{-1}(\Lambda p)\,\Lambda\,L(p)$ is a local Wigner rotation. It follows that $W^a{}_b(x) = W^a{}_b(\Lambda(x), p(x)) = \left[L^{-1}(\Lambda p)\,\Lambda\,L(p)\right]^a{}_b$, with a standard Lorentz transform (LT) $L^a{}_b(p)$ defined by,

$$\begin{aligned}L^0{}_0(p) &= \gamma,\\ L^0{}_i(p) = L^i{}_0(p) &= \frac{p^i}{m},\\ L^i{}_k(p) &= \delta_{ik} + (\gamma-1)\frac{p^ip^k}{|\overrightarrow{p}|^2},\end{aligned} \tag{19}$$